

Preliminary.

Def. A Topological space X is called T_4 if:

For any disjoint closed sets A and B ,
there exist disjoint open sets U and V such that $A \subseteq U$, $B \subseteq V$.

C - O chain Lemma.

A space X is T_4 if and only if

For any closed C and open U with $C \subseteq U$,
there exists an open set $\mathcal{O}$ such that $C \subseteq \mathcal{O} \subseteq \overline{\mathcal{O}} \subseteq U$.

Proof. Suppose that X is T_4 space. Given closed C and open U ,
 $C \subseteq U$ implies $C \cap (X \setminus U) = \emptyset$. That is, C and $X \setminus U$ are disjoint closed sets.
By T_4 condition, there exist disjoint open sets $\mathcal{O}, A$ such that

$$C \subseteq \mathcal{O}, X \setminus U \subseteq A, \mathcal{O} \cap A = \emptyset,$$

Now, $C \subseteq \mathcal{O} \subseteq \overline{\mathcal{O}}$, and $\mathcal{O} \cap A = \emptyset \Leftrightarrow \mathcal{O} \subseteq X \setminus A \Rightarrow \overline{\mathcal{O}} \subseteq X \setminus A$

Since the closure is the smallest closed set containing them, so that
 $X \setminus U \subseteq A \Leftrightarrow X \setminus A \subseteq U$ therefore

$$C \subseteq \mathcal{O} \subseteq \overline{\mathcal{O}} \subseteq U.$$

The converse can be shown for disjoint closed sets A and B ,
apply the chain for $A \subseteq X \setminus B$, similarly. $\square$

Note: $A \subseteq B \Leftrightarrow B^c \subseteq A^c$, $A \cap B = \emptyset \Leftrightarrow A \subseteq B^c \Leftrightarrow B \subseteq A^c$.

Def. Let X be a Topological space, and $A, B \subset X$ be disjoint closed sets. A real-valued Continuous map $f: X \rightarrow [a, b]$ is called Urysohn function:

$$f|_A = a \quad \text{and} \quad f|_B = b$$

Urysohn Lemma.

A T_4 -space has an Urysohn function for any two disjoint closed sets.

Tietze Extension Theorem.

Let X be a T_4 -space, and $A \subset X$ be a closed subset

For any continuous map $f: A \rightarrow \mathbb{R}$, there exists a continuous map

$g: X \rightarrow \mathbb{R}$ such that $g|_A = f$.